OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: CANADIAN SOLAR
MODEL / SERIES: HiKu CS3W-445MS
DOCUMENT TITLE: Canadian Solar HiKu High Power Mono PERC Module CS3W-445MS Datasheet
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1. PRODUCT IDENTIFICATION & ARCHITECTURE

Cell Technology:	Monocrystalline PERC
Module Construction:	Monofacial
Number of Cells / Format:	144 [2x(12x6)] Half-cut Mono PERC Cells
Glass / Front Encapsulant:	3.2 mm Tempered Glass with AR Coating
Frame Material:	Anodized Aluminium Alloy (Silver)
Junction Box / Protection:	IP68 with 3 bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)

Nominal Maximum Power (Pmax / Pnom):	445.0 Wp
Module Aperture Efficiency:	20.10 %
Open Circuit Voltage (Voc):	48.90 V
Short Circuit Current (Isc):	11.54 A
Maximum Power Voltage (Vmp):	40.90 V
Maximum Power Current (Imp):	10.89 A
Maximum System Voltage:	1500.0 V DC
Maximum Series Fuse Rating:	20.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS

Temp. Coefficient of Pmax (gamma):	-0.340 %/C
Temp. Coefficient of Voc (beta):	-0.260 %/C
Temp. Coefficient of Isc (alpha):	+0.050 %/C
Nominal Cell Temperature (NOCT/NMOT):	41 +/- 3 C (NMOT)
Operating Temperature Range:	-40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS

Length:	2108.0 mm
Width:	1048.0 mm
Thickness (Height):	35.0 mm
Gross Aperture Area:	2.2092 m2
Module Weight:	24.8 kg
Mechanical Design Load (Front):	5400 Pa (Snow load)
Mechanical Design Load (Rear):	2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE

Product Workmanship Warranty:	12.0 Years
Linear Power Output Warranty:	25.0 Years
Annual Linear Degradation:	0.55 % / year
End of Warranty Retained Output:	84.8 %

6. PROVENANCE & ARCHIVAL METADATA

Authoritative URL:	https://www.csisolar.com/wp-content/uploads/Canadian_Solar_Datasheet_HiK
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SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730